

CASE 46

LEPROMATOUS LEPROSY

T_H1 versus T_H2 responses in the outcome of infection.

Mature naive $CD4^+$ T cells emerging from the thymus can differentiate into effector $CD4^+$ T cells of several different phenotypes, each with different functions in the immune response. Two of these subsets are called T_H1 and T_H2 (Fig. 46.1). They develop from naive T cells activated by pathogen antigens in the presence of different signals provided by antigen-presenting cells and the local environment. These two types of effector T cells are distinguished chiefly by the cytokines that they secrete when they encounter their target cell. T_H1 cells secrete interleukin-2 (IL-2), interferon- γ (IFN- γ), and lymphotoxin (LT). T_H2 cells secrete IL-4, IL-5, IL-10,

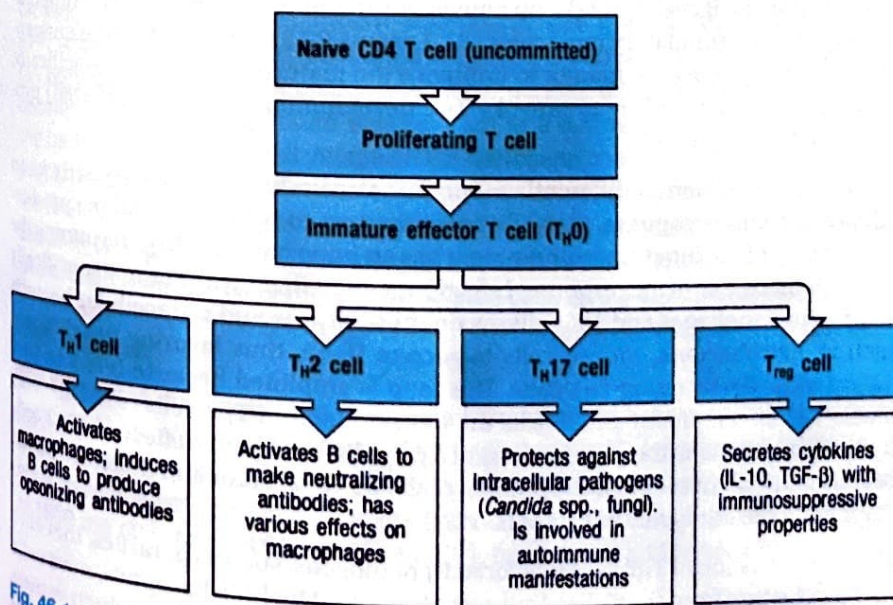


Fig. 46.1 $CD4^+$ T cells can differentiate into a variety of effector cells after encounter with antigen. Naive $CD4^+$ T cells first respond to their antigens (peptide:MHC class II complexes) by making IL-2 and proliferating. These cells then differentiate into a cell type sometimes known as T_{H0} . The T_{H0} cell has the potential to differentiate into distinct functional subsets (T_{H1} , T_{H2} , T_{H17} , and T_{reg} cells).

TOPICS BEARING ON THIS CASE:

Cytokine production in innate immunity

Differentiation of T_H1 versus T_H2 $CD4^+$ T cells

Role of cytokines in T-cell differentiation

Functions of T_H1 and T_H2 cells in immune responses

Responses to mycobacteria

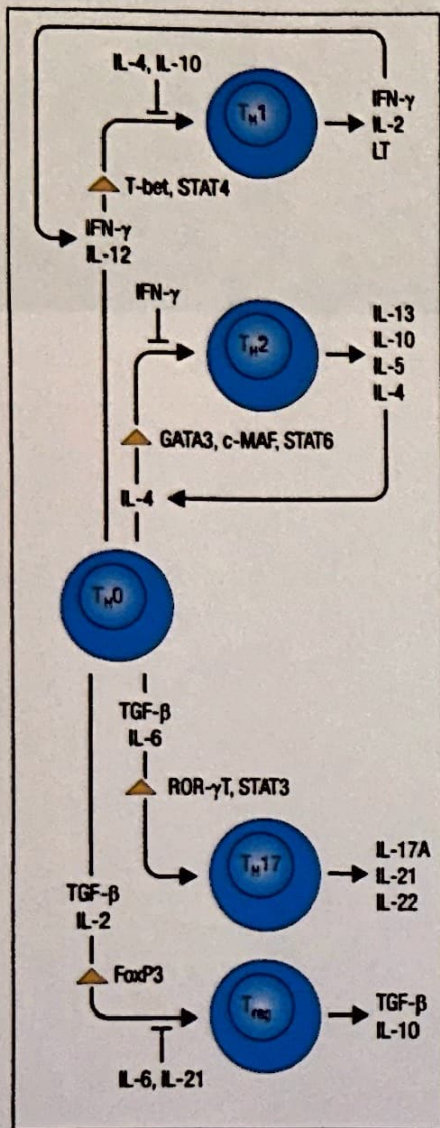


Fig. 46.2 Differentiation of T_H0 cells into T_H1 , T_H2 , T_H17 , or T_{reg} cells depends on the cytokines present. IL-4 induces T_H2 differentiation, whereas IL-12 and IFN- γ induce differentiation into T_H1 cells. TGF- β and IL-6 drive the differentiation of T_H17 cells, whereas TGF- β and IL-2 favor the generation of regulatory T (T_{reg}) cells. IL-4, which is secreted by T_H2 cells, also inhibits T_H1 development. Similarly, cytokines produced by T_H1 cells, for example IFN- γ , inhibit T_H2 differentiation. Development of T_H1 , T_H2 , T_H17 , and T_{reg} cells is driven by transcription factors induced by the cytokines. In particular, induction of the transcription factors T-bet and STAT4 in naive T cells leads to T_H1 development, and induction of GATA3 and STAT6 leads to T_H2 development, whereas induction of ROR- γ T and STAT3 favors the differentiation of T_H17 cells, and expression of FoxP3 leads to the development of T_{reg} cells.

and IL-13 (Fig. 46.2). $CD4^+$ T cells can also develop into effector T_H17 cells, which secrete IL-17A, or into induced regulatory T (T_{reg}) cells, each of which has distinct functional properties (see Cases 20 and 18, respectively).

The consequences of the decision to differentiate into T_H1 or T_H2 cells are profound. Selective production of T_H1 cells enables the immune response to activate macrophages and cell-mediated immunity, whereas selective production of T_H2 biases the response toward antibody production and promotes the differentiation of macrophages into alternatively activated type 2 macrophages. Which pathway a naive T cell will follow is decided during its first encounter with antigen and is dependent on cytokines. T_H1 differentiation is dependent on IL-12 and IFN- γ , whereas T_H2 differentiation is dependent on IL-4 (see Fig. 46.2). These cytokines trigger specific signal transduction pathways; for example, mice deficient in the intracellular signaling molecule STAT6, which is induced by IL-4, fail to develop T_H2 cells.

Other factors influencing the T-cell phenotype are the amount of antigen present and, thus, which cells are most likely to present it. Large amounts of antigen are usually presented by dendritic cells, which produce IL-12 and therefore favor T_H1 differentiation. A limited amount of antigen leads to preferential presentation by antigen-specific B cells that take up antigen more avidly; they induce T_H2 differentiation. The co-stimulatory molecules (B7.1 versus B7.2) expressed by the antigen-presenting cells are also thought to influence the maturation process: specifically, B7.1 (CD80) is more likely to provoke T_H1 development, while B7.2 (CD86) provokes T_H2 development.

Because the decision to differentiate into T_H1 versus T_H2 cells occurs early in an adaptive immune response, the ability of pathogens to stimulate cytokine production by cells of the innate immune system has an important role in determining the subsequent course of the response. Infectious agents that invade or nonspecifically activate macrophages and NK cells, as do most viruses and intracellular bacteria such as mycobacteria, induce cells to secrete IL-12, thus favoring the differentiation of IFN- γ -secreting T_H1 cells. This loop is amplified because IFN- γ in turn favors T_H1 development and blocks the development of T_H2 cells (see Fig. 46.2). IL-12 also enhances the proliferation of T_H1 cells but has no effect on T_H2 cells because they do not express the β chain of the IL-12 receptor and therefore do not respond to the mitogenic effects of IL-12.

The differentiation of T_H2 cells is favored by pathogens, such as parasites, that elicit IL-4 production from specialized subsets of cells that include basophils, mast cells, eosinophils, and thymus-derived invariant NKT cells (iNKT cells), which express both the NK1.1 marker and a T-cell receptor of restricted V_β and invariant V_α chain usage. This loop is amplified by the cytokines IL-4 and IL-10, produced by T_H2 cells. IL-4 promotes the development of T_H2 cells, and IL-10 blocks T_H1 development (see Fig. 46.2).

Once one T_H phenotype becomes dominant in the course of a response, it is difficult to shift the antigen-specific response to the other. One reason for this is that the cytokine products of T_H1 and T_H2 cells are reciprocally inhibitory (see Fig. 46.2). The outcome of certain infections is greatly influenced by the type of T-cell response elicited. As we see in this case, infection with *Mycobacterium leprae*, the leprosy bacillus, illustrates this phenomenon well.

The case of Ursula Iguaran: a T_H2 response to the leprosy bacillus has severe consequences.

Ursula first sought medical advice when she was 18 years old, having left her home in Colombia to attend Harvard University on a scholarship. From the age of 16 years, she had started to notice a gradual loss of sensation on the backs of her hands and had developed hypopigmented lesions over both arms. The lesions progressively became worse and she noticed that she was losing her eyelashes and hairs from her eyebrows. She also experienced recurrent nosebleeds. A month after first noticing the hair loss she decided to seek medical help.

On examination at the physician's office, Ursula seemed to be well, apart from her immediate symptoms. She reported a history of mild asthma, which required treatment with inhaled β_2 -adrenergic agents on an as-needed basis. Multiple hypopigmented macules (coin-sized raised lesions with ill-defined borders) were evident on her skin, along with cutaneous nodules 1 cm in diameter. These lesions were predominantly on her elbows, wrists, and hands (Fig. 46.3) and showed traces of dried blood; she also had similar lesions on her knees, ears, and buttocks. The absence of eyelashes and the ends of her eyebrows was obvious.

Cardiovascular and abdominal examinations were normal; a neurological examination was negative except for a decreased response to pinprick on the outer edges of the right and left hands and the right fourth and fifth fingers. There was a flexion contracture of the fourth and fifth fingers of both hands so that she could not straighten these fingers completely.

On blood test, Ursula's hematocrit was 35.1%; her white blood count was $7100 \mu\text{l}^{-1}$, with 68% neutrophils, 23% lymphocytes, 5% monocytes, and 4% eosinophils (all normal values). Serum electrolytes were normal. Because her symptoms were becoming more severe, Ursula was referred to a dermatologist. She told him that she had grown up in a small village on the Caribbean coast of Colombia where many people, including her mother, had leprosy. The dermatologist performed a biopsy of the lesions on her left arm and right forearm, which revealed numerous acid-fast bacilli in clumps. A routine hematoxylin and eosin stain of lesion tissue showed up numerous Virchow's cells (highly vacuolated cells of the macrophage lineage, also known as foam cells) and few lymphocytes (Fig. 46.4). Cultures for acid-fast bacilli were negative.

The suspected diagnosis of lepromatous leprosy led to a more extensive immunologic work-up. Delayed hypersensitivity skin tests with intradermal injections of candida, mumps, and tuberculin antigens showed no reactions when the injection sites were inspected 48 and 72 hours later. Ursula's serum IgG was mildly elevated at 1800 mg dl^{-1} (normal $600\text{--}1100 \text{ mg dl}^{-1}$); her IgA and IgM levels were normal.

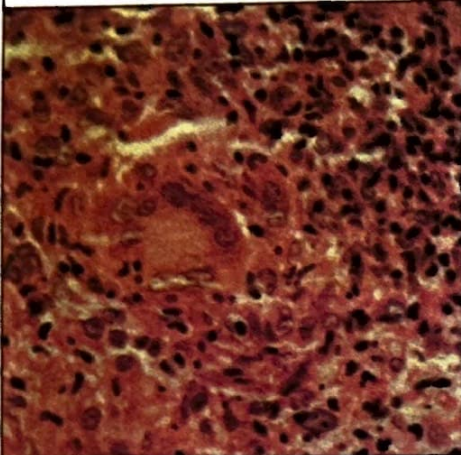
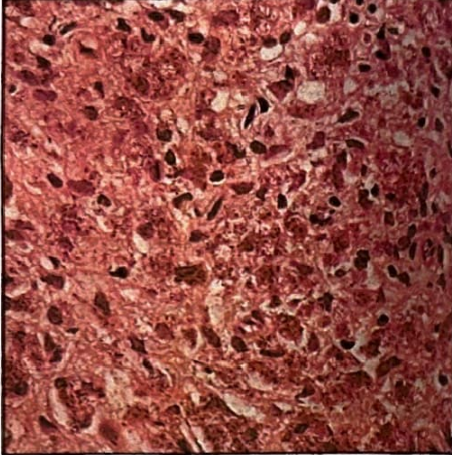
18-year-old female;
light-colored lesions on skin.

Acid-fast bacilli.
(? mycobacteria) in skin
lesions; leprosy?



Fig. 46.3 Cutaneous nodules in lepromatous leprosy. Patients with lepromatous leprosy have multiple skin lesions. This photograph shows subcutaneous nodules on the hand. Photograph courtesy of E. Gonzalez.

Fig. 46.4 Responses to *M. leprae* are sharply differentiated in lepromatous and tuberculoid leprosy. The photographs show sections of lesion biopsies stained with hematoxylin and eosin. Infection with *M. leprae* bacilli, which can be seen in the right-hand photograph as numerous small dark red dots inside macrophages, can lead to two very different forms of the disease. In tuberculoid leprosy (left), growth of the microorganism is well controlled by T_H1-like cells that activate infected macrophages. The tuberculoid lesion contains granulomas (see Case 26) and is inflamed, but the inflammation is localized and causes only local peripheral nerve damage. In lepromatous leprosy (right), infection is widely disseminated and the bacilli grow uncontrolled in macrophages. In the late stages there is severe damage to connective tissues and to the peripheral nervous system. There are several intermediate stages between these two polar forms. Photographs courtesy of G. Kaplan.

Infection with <i>Mycobacterium leprae</i> can result in different clinical forms of leprosy	
There are two polar forms, tuberculoid and lepromatous leprosy, but several intermediate forms also exist	
Tuberculoid leprosy	Lepromatous leprosy
	
Organisms present at low to undetectable levels	Organisms show florid growth in macrophages
Low infectivity	High infectivity
Granulomas and local inflammation. Peripheral nerve damage	Disseminated infection. Bone, cartilage, and diffuse nerve damage
Normal serum immunoglobulin levels	Hypergammaglobulinemia
Normal T-cell responsiveness. Specific response to <i>M. leprae</i> antigens	Low or absent T-cell responsiveness. No response to <i>M. leprae</i> antigens

A diagnosis of lepromatous leprosy was made on the basis of the presence of acid-fast bacilli in the biopsy and Ursula's progressive neurologic symptoms. She was placed on a multiple drug regime consisting of dapsone, clofazamine, and rifampin, drugs that kill *M. leprae*. Her skin lesions gradually flattened and improved.

Lepromatous leprosy.

The classical clinical feature in patients with leprosy is the association of cutaneous lesions, neuropathologic changes, and deformities. Leprosy is caused by *Mycobacterium leprae*, which colonizes macrophages and other host cells and multiplies within them. Mycobacteria within macrophages are protected from attack by antibody and can be eliminated only when their host macrophages are activated and produce increased amounts of nitric oxide, oxygen radicals, and other microbicidal molecules. *M. leprae* grows best at 30°C, and therefore lesions tend to appear on the extremities—the colder areas of the body—for example, the hands, ears, and buttocks as in Ursula's case. Unlike *M. tuberculosis*, *M. leprae* does not grow in culture.

The clinical symptoms of leprosy vary depending on the type of immune response to the mycobacteria. The clinical spectrum is typically divided into two polar forms, tuberculoid and lepromatous leprosy, although intermediate forms exist. Tuberculoid leprosy is associated with a vigorous T_H1-cell-mediated response against the bacillus. This results in macrophage activation with efficient killing of intracellular mycobacteria, localized tissue damage, and usually a milder clinical

picture. In the lepromatous form, the T_H1 cell-mediated response is defective and a T_H2 response predominates; this leads to a vigorous but ineffective antibody response against *M. leprae* and dissemination of the bacilli to other sites in the body, which results in further tissue destruction and aggravation of the symptoms. The importance of T_H1 -derived IFN- γ in containing mycobacterial infections is further illustrated by the observation that infants with genetic defects in IFN- γ production or the IFN- γ receptor die from disseminated mycobacterial infections (see Case 24).

Infection with *M. leprae* illustrates a situation in which the same microorganism, in different individuals, can trigger either a T_H1 or a T_H2 response. A T_H1 response predominates in tuberculoid leprosy, in which the mycobacteria are contained within well-circumscribed granulomas and propagate poorly, usually accompanied by subsequent minimal tissue damage. In contrast, a T_H2 response predominates in lepromatous leprosy, in which the mycobacteria propagate rapidly, with resulting extensive tissue damage. Analysis of mRNA isolated from lesions of patients with lepromatous and tuberculoid leprosy illustrates the cytokine patterns in the two forms of the disease. T_H2 cytokines (IL-4, IL-5, and IL-10) dominate in the lepromatous form, whereas T_H1 cytokines (IL-2, IFN- γ , and LT) dominate in the tuberculoid form (Fig. 46.5).

The neurologic damage in leprosy has two main causes. It can arise from bacterial multiplication within Schwann cells—the cells that form the insulating myelin sheath around some nerve cell axons. Disruption of the myelin sheath interferes with the normal conduction of nerve impulses along the axon. In the tuberculoid form, nerve damage also arises from the formation of granulomas and inflammation of the tissue surrounding the nerve. The nerve damage results in dysfunctional nerve terminals, causing decreased sensation and eventually a loss of motor function.

The nosebleeds that Ursula experienced are common in leprosy. They are due to large numbers of *M. leprae* in the nasal tissue with extensive involvement of the nasal mucosa, leading to congestion and breakage of blood vessels.

The T_H2 response can influence the course of the infection in various other ways. By binding to mycobacterial antigens displayed on the surface of infected cells, antibodies against the leprosy bacillus can interfere with the action of cytotoxic $CD8^+$ T cells. In addition to their cytolytic function, $CD8^+$ T cells also respond to antigen by secreting cytokines. Patients with lepromatous leprosy have $CD8^+$ T cells that suppress the T_H1 response by making IL-10 and LT. IL-10 inhibits the development of T_H1 cells and inhibits both cytokine release from macrophages and their capacity to kill internalized microorganisms. LT also inhibits the intracellular killing capacity of macrophages. Inhibition of macrophages leads to decreased production

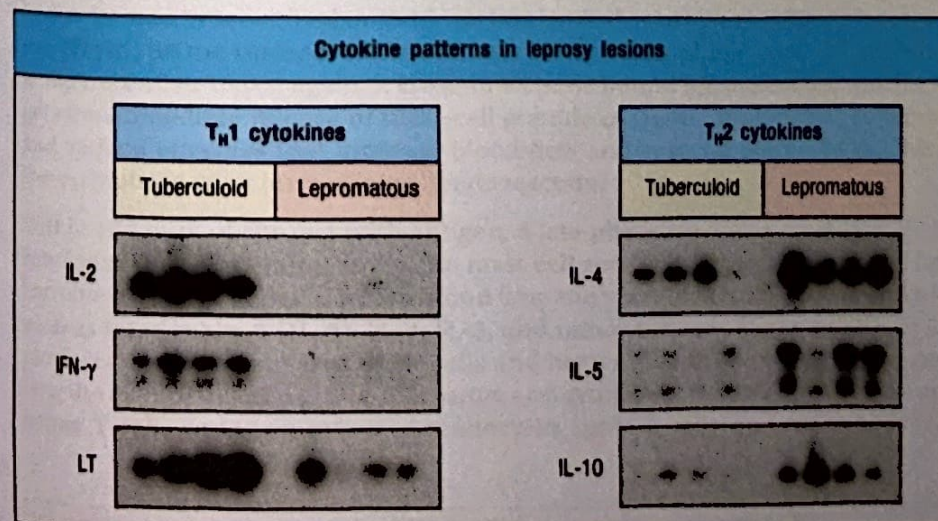


Fig. 46.5 Cytokine patterns in leprosy lesions. The cytokine patterns in the two polar forms of leprosy are distinctly different, as shown by northern blot analysis of the mRNA from lesions of three patients with lepromatous leprosy and three patients with tuberculoid leprosy. Cytokine mRNAs typically produced by T_H2 cells predominate in the lepromatous form, whereas cytokines produced by T_H1 cells predominate in the tuberculoid form. Cytokine blots courtesy of R.L. Modlin.

of IL-12, fewer T_H1 cells, and more T_H2 cells. In contrast, patients with the less destructive tuberculoid leprosy lack suppressor $CD8^+$ T cells and thus make a vigorous T_H1 response, leading to macrophage activation and the destruction of the leprosy bacilli.

Questions.

- 1 Ursula did not respond to candida and mumps antigens, which are common recall antigens, with a delayed-type hypersensitivity reaction. Give a possible explanation.
- 2 Which cytokine might be beneficial to a patient with lepromatous leprosy?
- 3 Describe the mechanism for Ursula's hypergammaglobulinemia.
- 4 Why would Ursula be prone to asthma?